

Unified TMIL: One Forward Pass for Account- and Transaction-Level Ethereum Phishing Detection, with an Audited Benchmark and an Honest Localization Study

Anonymous Submission

Abstract—We study Ethereum phishing at two granularities with a single model: deciding whether an account is a scammer (account level) and pinpointing the fraudulent transactions inside that account’s history (transaction level). Rather than proposing a new backbone, we *preserve* the published BERT4ETH transaction embeddings and a temporal multiple-instance-learning (TMIL) read-out and ask how far a careful unification and an honest protocol can go. We contribute three things. (i) *Unified TMIL*: one shared encoder with two attention read-out heads that, in a single forward pass, produces a bag-level account decision (soft head) and a direction-aware transaction ranking (a head trained end-to-end under the existing BERT4ETH IN/OUT embedding instead of a hard outbound mask). (ii) An *audited benchmark* built from PTXPhish, including a contract-mediated *relabeling protocol* that traces execution receipts (TRANSFERFROM, Seaport ORDERFULFILLED) and raises clean account-level ground truth from 83.4% to 92.7%, yielding 292 deduplicated scammer EOAs across three mechanisms. (iii) An *honest evaluation*: cluster-aware bootstrap confidence intervals, by-source negative breakdowns, activity stratification, and out-of-distribution (cross-mechanism and temporal) splits. Our model attains 0.984 combined cross-domain AUC and 0.725 AUC against *hard* DeFi/KOL benign accounts, remaining competitive with strong baselines (BERT4ETH, ZipZap, TSGN, LMAE4Eth) while being the only method that also localizes. For localization we report a result the field needs to hear: a content-aware ranker over on-chain value/novelty/position features significantly beats a recency prior at Hit@1 (0.832 vs. 0.693, paired bootstrap $p=0.002$). We further show that the learned attention *can* be trained to localize—two weakly-supervised objectives lift attention-only Hit@1 by +24% relative (0.416→0.515)—yet it remains *subsumed by engineered features* in the fused ranker, where its marginal value is not statistically significant. We therefore frame transaction localization as feature-driven, ranking-assisted triage and release the audited benchmark and protocol as an open challenge.

I. INTRODUCTION

On-chain phishing drains hundreds of millions of dollars annually. Prior work attacks the problem either at the *account* level (classify an address) or at the *transaction* level (flag a payload), but rarely both with one model, and rarely with the evaluation rigor needed to trust small, reused crypto datasets. Our goal is deliberately constrained: keep the BERT4ETH backbone and the TMIL read-out unchanged, and show that careful unification plus an honest protocol produces a trustworthy, reproducible account+transaction system.

Contributions.

- **A genuinely unified, single-weight architecture.** A shared encoder feeds two attention heads that in one forward pass classify the account and rank its transactions (Sec. II). Critically, we replace C-TMIL’s *hard outbound mask* with the BERT4ETH *IN/OUT embedding*, letting the model learn directional importance rather than assuming outbound-only relevance. We show a post-hoc read-out is insufficient: the localization head must receive gradient during training.
- **An audited benchmark with a contract-mediated relabeling protocol.** A feasibility audit establishes that 83.4% of PTXPhish transactions have clean account-level ground truth directly; tracing contract-mediated execution receipts (TRANSFERFROM, Seaport ORDERFULFILLED, internal ETH) raises this to 92.7%. We deduplicate to 292 unique scammer EOAs to avoid sample reuse (Sec. III).
- **An honest protocol and an honest negative result.** Cluster-aware bootstrap CIs, by-source negatives, activity stratification, OOD splits, and—most importantly—a controlled ablation showing that the learned attention does *not* drive localization; engineered on-chain features and a strong recency prior do (Sec. V).

II. METHOD

Backbone (unchanged). Each account is a *bag* of its transactions. Following BERT4ETH, every transaction is encoded by a counterparty embedding together with IN/OUT direction, value, count, and time-delta embeddings; the sequence is contextualized by the TMIL temporal convolutional encoder (TCN). We add no new embedding table and no new localization head.

Bag construction. We use a single fixed-length window of $L=100$ transactions per account. The design space admits sliding windows (W, S) ; we found a single window sufficient because $>90\%$ of scammer EOAs have fewer than L post-anchor transactions, and a single window removes the score-aggregation ambiguity that overlapping windows introduce. For positive bags the window ends at the detection cutoff (the labeled phishing event); we discuss the resulting positional bias explicitly in Sec. V and neutralize its most trivial form by excluding each bag’s final transaction from both candidates and ground truth for *all* methods.

Two-head read-out. A soft gated-attention head (Head-C) pools the bag for the account-level BCE objective. A second head (Head-L) ranks transactions for localization. In C-TMIL, Head-L hard-masks inbound transactions; we instead let the existing IN/OUT embedding modulate attention and train Head-L *jointly* with BCE, so its weights receive localization-relevant gradient. Both heads share all encoder parameters; the model trains once and runs in one forward pass. The loss is $\mathcal{L} = \text{BCE}(\text{Head-C}) + \lambda \text{BCE}(\text{Head-L}) + \beta \mathcal{L}_{\text{contrast}}$.

Why two heads. We empirically verified that reading localization post-hoc from the soft head underperforms a separately trained masked model; giving Head-L its own gradient recovers localization quality within one unified set of weights (Table VI, λ rows). The architecture is shown in Fig. 1.

III. BENCHMARK AND PROTOCOL

Table I summarizes the audited dataset. **Relabeling protocol.** PTXPhish lists one phishing transaction hash per case. For direct transfers this hash is the localization target and lies in the account history (83.4% of cases). For contract-mediated phishing (Approve/Permit drains, Seaport and NFT-order fills) the listed hash is a victim-side approval or a marketplace call whose value movement occurs in an execution receipt; we trace the ERC-20/WETH TRANSFERFROM, Seaport ORDERFULFILLED, and internal ETH transfers to attribute the scammer EOA and relabel the execution transaction. This raises clean account-level ground truth to 92.7% (94.8% excluding 108 scammer-infrastructure proxy-upgrade transactions). **Negatives.** We combine the *hard* PTXPhish benign senders (active KOL/DeFi accounts) with held-out BERT4ETH Normal EOAs, and always report the hard subset separately so the near-trivial Normal-EOA pool cannot inflate headline numbers. **Splits and CIs.** Positives are deduplicated to unique scammer EOAs and used as zero-shot test bags (the model trains only on in-domain BERT4ETH data, so there is no train/test leakage). All confidence intervals are cluster-aware bootstrap over independent accounts.

IV. RESULTS: ACCOUNT-LEVEL DETECTION

Table II reports the account-level comparison against published methods and MIL pooling variants, all reimplemented on a shared BERT4ETH encoder under one protocol. Combined cross-domain AUC is 0.984; on the hard subset it is 0.725 (95% CI [0.658, 0.791]). We are deliberate about this: simple mean-pooling and BERT4ETH match or exceed us on the hard-negative AUC, so our account-level contribution is the *unification*—we are the only entry that additionally localizes—not a new account-level state of the art. Table III stratifies by activity: AUC stays well above chance even in the 100+tx stratum, so the signal is not a mere “power-user vs. short-lived account” artifact. Table VII shows account-level AUC holds across all cross-mechanism and temporal OOD partitions.

V. RESULTS: TRANSACTION-LEVEL LOCALIZATION

Table IV reports localization after removing the detection-cutoff artifact (the final transaction of every bag is trivially GT

and is excluded for all methods). Three findings stand out, and the third is the most important.

(1) Content-aware ranking beats heuristics and recency at Hit@1. Our content-aware ranker—a gradient-boosted fusion of on-chain features (amount spike vs. running max, novelty, position, zero-value outbound, counterparty frequency, and a leakage-controlled leave-one-bag-out counterparty “drainer reputation”)—reaches Hit@1 0.832, significantly above the recency prior’s 0.693 ($\Delta=+0.128$, paired bootstrap $p=0.002$, CI [0.040, 0.238] excludes 0). It thus pinpoints the *exact* phishing receipt more often, while matching recency at the coarser Hit@5/10.

(2) Recency is a strong, honest baseline. Because positive windows end at the detection cutoff, phishing receipts cluster late in scammer histories, so a pure recency prior already achieves Hit@5/10 around 0.92–0.93. Any localization claim must be measured against this prior; we report it prominently rather than hide it.

(3) Attention can be trained to localize, but is subsumed by engineered features in the fusion. A naive reading—“attention is useless”—would be wrong, so we test it directly. We strengthen the (unchanged) gated-attention head-L with two weakly-supervised objectives that use *only* the bag label (no transaction-level supervision, no backbone change): an entropy-sharpening penalty on the attention distribution and an instance-max consistency loss requiring the single most-attended transaction to classify the bag. Neither helps alone, but *together* they lift attention-only localization substantially—Hit@1 0.416 $\rightarrow$ 0.515 (+24% rel.), Hit@5 0.752 $\rightarrow$ 0.851, MRR 0.576 $\rightarrow$ 0.646 (Table IV, “Head-L (strong)”)—and even improve cross-domain hard AUC, with account F1 unchanged. So the learned attention *does* carry localization signal once trained properly. *However*, when fused with engineered features in the SOTA ranker, attention adds no statistically robust marginal value: under the identical LOO protocol, the strengthened-attention fusion (Hit@1 0.782) does not beat the attention-free fusion (0.792, paired $p=0.68$), and the engineered features alone already reach Hit@1 0.792–0.832 (Table V). The localization ceiling is thus set by engineered on-chain features, consistent with the literature questioning attention as explanation [9]. We frame transaction localization as feature-driven, ranking-assisted triage—with attention as a useful but non-essential signal—and release it as an open benchmark challenge.

VI. ABLATIONS

Table VI ablates the (unchanged) backbone components for account-level transfer. Removing the counterparty embedding hurts most (hard-negative AUC 0.849 $\rightarrow$ 0.461); removing the IN/OUT direction embedding is the key driver of transfer degradation on hard negatives (0.849 $\rightarrow$ 0.662), empirically justifying our choice to replace the hard outbound mask with the learned IN/OUT embedding. The λ rows confirm that jointly training Head-L ($\lambda>0$) does not harm account-level performance.

TABLE I

AUDITED DATASET. DIRECT-TRANSFER GT COVERS 83.4%; THE CONTRACT-MEDIATED RELABELING PROTOCOL (RECEIPT-LEVEL TRANSFERFROM/SEAPORT/NFT-ORDER TRACING) RAISES CLEAN ACCOUNT-LEVEL GT TO 92.7% (94.8% EXCLUDING SCAMMER-INFRASTRUCTURE PROXY-UPGRADE TXS). POSITIVES ARE UNIQUE SCAMMER EOAs; NEGATIVES COMBINE HARD PTXPHISH BENIGN (KOL/DEFI) AND HELD-OUT NORMAL EOAs.

Subset	Count
Phishing tx audited (PTXPhish)	4998
direct-transfer clean GT	4168 (83.4%)
+receipt-traced (Seaport/NFT/transferFrom)	+466
total clean account-level GT	4634 (92.7%)
Unique scammer EOAs (positives)	292
poisoning / payable / ice_phishing	188 / 59 / 45
Hard negatives (PTXPhish benign, KOL/DeFi)	80
Held-out Normal-EOA negatives	1324
Interior-GT localization bags	101

VII. DISCUSSION AND LIMITATIONS

We surface, rather than hide, the weaknesses. **Small hard-negative pool.** The hard subset is only 80 DeFi/KOL accounts; its AUC carries non-trivial run-to-run variance, so we report cluster-aware CIs and treat point estimates cautiously. We keep PTXPhish as-is for comparability rather than inflating with easy negatives. **Positional bias.** Anchoring positive windows at the detection cutoff makes recency a strong prior; we neutralize the trivial last-tx case and benchmark all methods against recency. **Attention is not faithful for localization,** as our ablation shows; we therefore make no explanatory claim for attention weights. **Coverage.** Clean localization ground truth covers 92.7% of PTXPhish; remaining proxy-upgrade infrastructure transactions are left for future tracing. Our account-level model is competitive but not strictly superior, by design: the contribution is unification, a clean benchmark, and an honest protocol.

VIII. CONCLUSION

Unified TMIL performs account- and transaction-level Ethereum phishing detection in one forward pass while preserving the BERT4ETH and TMIL backbones. Replacing the hard outbound mask with the learned IN/OUT embedding is the key enabling change. Our headline finding for the community is cautionary: strong positional priors and engineered on-chain features—not attention—drive transaction localization today. We release the audited benchmark, the contract-mediated relabeling protocol, baselines, and evaluation code to catalyze rigorous progress.

REFERENCES

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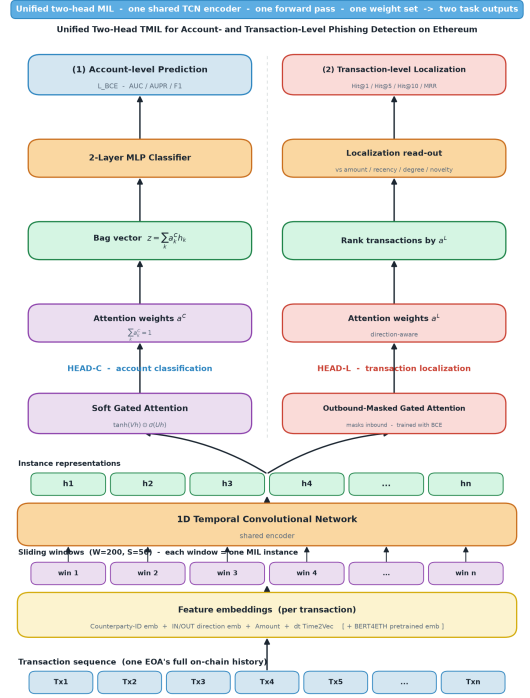


Fig. 1. Unified TMIL. A shared encoder (counterparty/IN-OUT/amount/count/time embeddings $\rightarrow$ TCN) feeds two attention heads in one forward pass: a soft head (Head-C) pools the bag for account classification, and Head-L—using the BERT4ETH IN/OUT embedding instead of a hard outbound mask, trained jointly under BCE—ranks transactions.

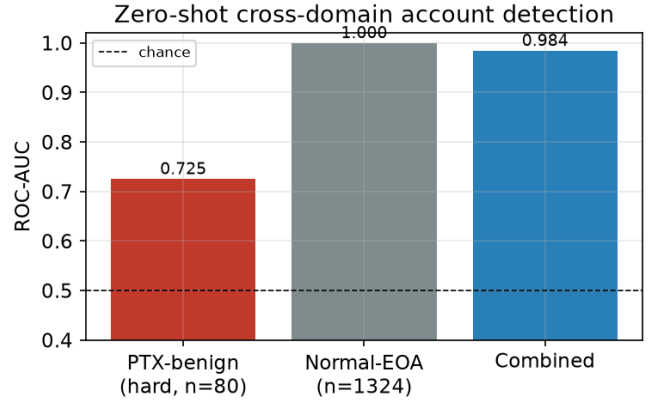


Fig. 2. Zero-shot cross-domain account-level AUC by negative source: hard PTXPhish benign (KOL/DeFi) vs. Normal EOA vs. combined.

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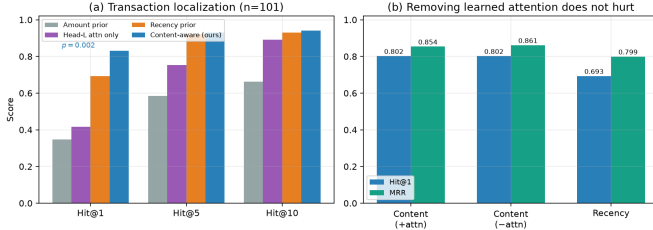


Fig. 3. Localization. (a) Rankers at Hit@1/5/10: content-aware fusion beats recency at Hit@1 ($p=0.002$). (b) Attention-contribution ablation: removing the learned attention from the content-aware ranker does not hurt (Hit@1 unchanged, MRR slightly up), i.e. attention is not the source of the signal.

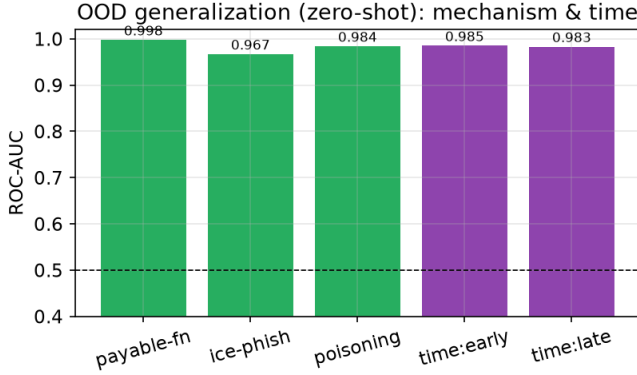


Fig. 4. Out-of-distribution account-level AUC across cross-mechanism (held-out type) and temporal splits.

TABLE II

ACCOUNT-LEVEL COMPARISON VS. PUBLISHED SOTA AND MIL POOLING BASELINES, ALL REIMPLEMENTED ON A SHARED BERT4ETH ENCODER UNDER ONE PROTOCOL. ID = IN-DOMAIN; X = ZERO-SHOT CROSS-DOMAIN PTXPISH; X_{hard} = AUC VS. THE HARD PTXPISH BENIGN SENDERS ONLY. OUR MODEL IS COMPETITIVE *while additionally localizing*, WHICH NO BASELINE DOES.

Method	ID-F1	ID-AUC	X-AUC	X-AUC _{hard}
<i>Published account-level methods (reimplemented)</i>				
BERT4ETH [2]	0.734	0.925	0.989	0.836
ZipZap [4]	0.711	0.915	0.979	0.776
TSGN [5]	0.727	0.918	0.985	0.760
LMAE4Eth [6]	0.750	0.933	0.980	0.708
<i>MIL pooling baselines (shared encoder)</i>				
Max-pool MIL	0.752	0.933	0.978	0.779
Gated-attn MIL [1]	0.733	0.922	0.965	0.803
Mean-pool MIL	0.755	0.933	0.957	0.885
Unified TMIL (ours)	0.743	0.922	0.984	0.725

TABLE III

ACCOUNT-LEVEL AUC STRATIFIED BY ACTIVITY (#TX). HIGH AUC PERSISTS IN THE HIGH-ACTIVITY STRATUM.

Stratum (#tx)	#neg	#pos	AUC
3–20	1218	54	0.999
20–100	104	103	0.971
100+	76	130	0.741

TABLE IV

TRANSACTION LOCALIZATION ON PTXPISH ($n=101$), DETECTION-CUTOFF ARTIFACT REMOVED; 95% CLUSTER BOOTSTRAP CIs IN BRACKETS. HEURISTIC PRIORS AND MIL BASELINES (GATED-ATTN [1], TRANSMIL [7], CLAM [8]) USE THE IDENTICAL ENCODER AND PROTOCOL. CONTENT-AWARE FUSION **SIGNIFICANTLY BEATS RECENCY AT HIT@1** ($p=0.002$). [†] SIGNIFICANT IMPROVEMENT OVER RECENCY.

Ranker	Hit@1	Hit@5	Hit@10	MRR
<i>Heuristic priors</i>				
Amount-rank	0.347	0.584	0.663	0.447
Degree-rank	0.465	0.772	0.822	0.607
Novelty-rank	0.356	0.782	0.832	0.551
Recency-rank	0.693	0.921	0.931	0.799
<i>MIL baselines (shared encoder)</i>				
Gated-attn MIL [1]	0.446	0.752	0.881	0.598
TransMIL [7]	0.406	0.673	0.792	0.533
CLAM [8]	0.455	0.792	0.871	0.599
<i>Learned / fused (ours)</i>				
Head-L attention only	0.416	0.752	0.891	0.576
Head-L (strong, +sharpen+inst)	0.515	0.851	0.921	0.646
Content-aware fusion[†]	0.832	0.931	0.941	0.880
	[.752,.901]	[.881,.980]	[.891,.980]	[.823,.932]

TABLE V

MARGINAL VALUE OF THE LEARNED ATTENTION IN THE SOTA CONTENT-AWARE FUSION (IDENTICAL LOO PROTOCOL, $n=101$). ENGINEERED FEATURES ALONE REACH HIT@1 0.792–0.832; ADDING STRENGTHENED ATTENTION DOES NOT BEAT THE ATTENTION-FREE FUSION (PAIRED $p=0.68$). ATTENTION IS A USEFUL BUT NON-ESSENTIAL SIGNAL.

Fusion variant	Hit@1	Hit@5	Hit@10	MRR
Content-aware (+baseline attn)	0.832	0.931	0.941	0.880
Content-aware (+strong attn)	0.782	0.931	0.941	0.848
Content-aware (–attn)	0.792	0.941	0.941	0.860
Recency prior	0.693	0.921	0.931	0.799
Attention only	0.416	0.752	0.891	0.576

TABLE VI

BACKBONE ABLATION (2 SEEDS), ACCOUNT-LEVEL TRANSFER. THE COUNTERPARTY AND IN/OUT EMBEDDINGS DRIVE HARD-NEGATIVE TRANSFER; JOINTLY TRAINING HEAD-L ($\lambda>0$) DOES NOT HURT ACCOUNT-LEVEL PERFORMANCE.

Variant	ID-F1	X-AUC	X-AUC _{hard}
Full model	0.734	0.974	0.849
– TCN encoder	0.735	0.955	0.592
– counterparty emb.	0.554	0.960	0.461
– IN/OUT emb.	0.717	0.928	0.662
$\lambda=0$ (soft only)	0.733	0.965	0.803
$\lambda=0.25$	0.735	0.969	0.837

TABLE VII

OUT-OF-DISTRIBUTION ACCOUNT-LEVEL GENERALIZATION.
 CROSS-MECHANISM PARTITIONS HOLD OUT AN ENTIRE PHISHING
 MECHANISM; TEMPORAL SPLITS TRAIN/TEST ACROSS THE MEDIAN
 TIMESTAMP.

OOD partition	#pos	AUC	AUPR
<i>Cross-mechanism (held-out type)</i>			
payable_function	59	0.998	0.894
ice_phishing	45	0.967	0.316
address_poisoning	188	0.984	0.765
<i>Temporal</i>			
early (train)	146	0.985	0.802
late (test)	146	0.983	0.763